Tao Wen

CONTACT INFORMATION

Decision and Cognitive Sciences Research Centre
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RESEARCH INTERESTS

Complexity Science, Network Science, Game Theory, Fractal Property, Group Decision-making

EDUCATION

University of Manchester — Manchester, UK
Integrated PhD and MSc in Data Analytics and Society (ESRC funded) — Expect Sep. 2024
Advisor: Professor Yu-wang Chen and Tahir Abbas Syed

Singapore University of Technology and Design (SUTD) — Singapore
MEng. (Research) in Science, Mathematics & Technology — Sep. 2021
Advisor: Professor Kang Hao Cheong

Northwestern Polytechnical University (NWPU) — Xi'an, China
B. S. in *Detection, Guidance and Control Technology (System Engineering)* — June 2019
Advisor: Professor Wen Jiang and Yong Deng

PUBLICATIONS
[Google Scholar]
Citation: 405
H-Index: 12
I10-Index: 12

* Corresponding Author; † Contribute equally.

2022

[1] **Tao Wen**, Kang Hao Cheong*, Joel Weijia Lai, Jin Ming Koh, Eugene V. Koonin. "Extending the lifespan of multicellular organisms via periodic or stochastic intercellular competition," *Physical Review Letters*. vol. 128, p. 218101, 2022. (**The Highest Impact Letter in Physics**) **Impact Factor: 9.185**

[2] Kang Hao Cheong*†, **Tao Wen**†, Sean Benler, Jin Ming Koh, Eugene V. Koonin*. "Alternating lysis and lysogeny is a winning strategy in bacteriophages due to Parrondo's Paradox," *Proceedings of the National Academy of Sciences*. vol. 119, p. e2115145119, 2022. **Impact Factor: 12.779**

[3] **Tao Wen**, Qiuya Gao, Yu-wang Chen, Kang Hao Cheong*. "Exploring the Vulnerability of Logistics Networks by Entropy: An Asia-Europe Maritime Transportation Network Case Study," *Reliability Engineering & System Safety*. Accepted. **Impact Factor: 7.247**

[4] **Tao Wen**, Qiuya Gao, Tamás Kalmár-Nagy, Yong Deng*, Kang Hao Cheong*. "A Review of Predator-Prey Systems with Dormancy of Predators," *Nonlinear Dynamics*. vol. 107, pp. 3271-3289, 2022.

[5] **Tao Wen**, Joel Weijia Lai, Kang Hao Cheong*. "Switching between two losing stocks may enable paradoxical win: an empirical analysis," *Fractals*. Accepted.

2021

[6] **Tao Wen**, Jinde Cao, Kang Hao Cheong*. "Gravity-based Community Vulnerability Evaluation Model in Social Networks: GBCVE," *IEEE Transactions on Cybernetics*. Accepted. **Impact Factor: 19.118**

[7] **Tao Wen**, Kang Hao Cheong*. "The Fractal Dimension of Complex Networks: A Review," *Information Fusion*. vol. 73, pp. 87-102, 2021. **Impact Factor: 17.564**

[8] **Tao Wen**, Eugene V. Koonin, Kang Hao Cheong*. "An alternating active-dormitive strategy enables disadvantaged prey to outcompete the perennially active prey through Parrondo's paradox," *BMC Biology*. vol. 19, p. 168, 2021. **Impact Factor: 7.364**

[9] Qiuya Gao, **Tao Wen**, Yong Deng*. "Information Volume Fractal Dimension". *Fractals*. vol. 29, p. 2150263, 2021.

[10] Yayun Dai, Guangjie Zhan, Ye Ye, Wei Bao, **Tao Wen**, Kang Hao Cheong*, Nenggang Xie*. "Game dynamics of emotion evolution based on the Moran process," *Chaos: An Interdisciplinary Journal of*

Nonlinear Science. vol. 31, p. 033153, 2021.

[11] Prabal Datta Barua, Nadia Fareeda Muhammad Gowdh, Kartini Rahmat, Norlisah Ramli, Wei Lin Ng, Wai Yee Chan, Mutlu Kuluozturk, Sengul Dogan*, Mehmet Baygin, Orhan Yaman, Turker Tuncer, **Tao Wen**, Kang Hao Cheong*, U. Rajendra Acharya. "Automatic COVID19 detection using exemplar hybrid deep features with X-ray images," *International Journal of Environmental Research and Public Health*. vol. 18, p. 8052, 2021.

2020

[12] Kang Hao Cheong*, **Tao Wen**, Joel Weijia Lai. "Relieving cost of epidemic by Parrondo's paradox: A COVID-19 case study," *Advanced Science*. vol. 7, p. 2002324, 2020. [Press] **Impact Factor: 17.521**

[13] **Tao Wen**, Danilo Pelus, Yong Deng*. "Vital Spreaders Identification in Complex Networks with Multi-Local Dimension," *Knowledge-Based Systems*. vol. 195, p. 105717, 2020. **Impact Factor: 8.139**

[14] **Tao Wen**, Yong Deng*. "The vulnerability of community structure in complex networks: An entropy approach," *Reliability Engineering & System Safety*. vol. 196, p. 106782, 2020. (**ESI Highly Cited Paper**) **Impact Factor: 7.247**

[15] **Tao Wen**, Yong Deng*. "Identification of influencers in complex network by local information dimensionality," *Information Sciences*. vol. 512, pp. 549-562, 2020. (**ESI Highly Cited Paper**) **Impact Factor: 8.233**

2019

[16] **Tao Wen**, Shuyu Duan, Wen Jiang*. "Node similarity measuring in complex networks with relative entropy," *Communications in Nonlinear Science and Numerical Simulation*. vol. 78, p. 104867, 2019.

[17] **Tao Wen**, Wen Jiang*. "Identifying influential nodes based on fuzzy local dimension in complex networks," *Chaos, Solitons & Fractals*, vol. 119, pp. 332-342, 2019. **Impact Factor: 9.922**

[18] **Tao Wen**, Wen Jiang*. "Measuring the complexity of complex network by Tsallis entropy," *Physica A: Statistical Mechanics and Its Applications*. vol. 526, p. 121054, 2019.

[19] Shuyu Duan, **Tao Wen**, Wen Jiang*. "A new information dimension of complex network based on Rényi entropy," *Physica A: Statistical Mechanics and Its Applications*, vol. 516, pp. 529–542, 2019.

2018

[20] **Tao Wen**, Moxian Song, Wen Jiang*. "Evaluating topological vulnerability based on fuzzy fractal dimension," *International Journal of Fuzzy Systems*, vol. 20, pp. 1956–1967, 2018.

[21] **Tao Wen**, Wen Jiang*. "An information dimension of weighted complex networks," *Physica A: Statistical Mechanics and Its Applications*, vol. 501, pp. 388 – 399, 2018.

Conference paper

[22] **Tao Wen**, Yu-wang Chen*, Tahir Abbas Syed. "Ranking Influential Users in Social Networks by the Evidential Reasoning Algorithm," *26th International Conference on Multiple Criteria Decision Making (MCDM2022)*. Portsmouth, UK, June 2022. (Abstract)

[23] **Tao Wen**, Shuyu Duan, Wen Jiang*. "Forecasting time series based on visibility graph and relative entropy," *The Ninth Chinese Information Fusion Conference*, Taiyuan, China, Oct. 2019. (In Chinese)

[24] Shuyu Duan, **Tao Wen**, Xinyang Deng, Wen Jiang*. "Identifying influential nodes based on Tsallis entropy and information dimension," *The Ninth Chinese Information Fusion Conference*, Taiyuan, China, Oct. 2019. (In Chinese)

[25] Shuai Xu, Zichang He, **Tao Wen**, Wen Jiang*. "A Physarum-inspired Model for the Path Planning of Uninhabited Combat Air Vehicle," *The Eighth Chinese Information Fusion Conference*, Xi'an, China, July 2018. (In Chinese)

Paper submitted to Journal

[26] **Tao Wen**, Kang Hao Cheong*. "Analyzing and Predicting Time Series by Visibility Graph: A Review," *TBD*.

[27] **Tao Wen**, Eugene V. Koonin, Kang Hao Cheong*. "Paradoxical Winning of Host-parasite Evolutionary Dynamics by Parrondo's Paradox," *TBD*.

[28] Zeyi Liu†, **Tao Wen**†, Yong Deng*, Hamido Fujita. "Cooperation-guided Experts Importance

Identification Model with Fuzzy Framework: A Network Design," *Information Fusion*.

[29] Qiuya Gao, **Tao Wen**, Xiangfeng Dai, Mengge Chai, Kang Hao Cheong*, Zhen Wang*. "Measuring Similar Nodes in Complex Networks based on Gravity Model," *Complex & Intelligent Systems*.

[30] Qiuya Gao, **Tao Wen**, Yong Deng*. "A Novel Network-based and Divergence-based Time Series Forecasting Method," *Information Sciences*.

[31] Jie Zhao, **Tao Wen**, Hadi Jahanshahi, Kang Hao Cheong*. "The random walk based gravity model to identify influential nodes in complex networks," *Information Sciences*.

[32] Kang Hao Cheong*, Michael C. Jones, **Tao Wen**. "'Governing' the ungovernable: When will we learn?" *TBD*.

[33] **Tao Wen**, Yong Deng*. "Information Volume Multifractal Dimension". Prepare to submit.

RESEARCH EXPERIENCE

Data Analytics for Addressing Fake News and Deepfakes in Social Networks Sep. 2021–Now
University of Manchester, Advisor: Professor Yu-wang Chen and Tahir Abbas Syed

Analyzing the Parrondo's paradox phenomenon in complex networks Jan. 2020–Sep. 2021
Singapore University of Technology and Design, Advisor: Professor Kang Hao Cheong

- Explored the increase in economic and health cost caused by different strategies (open community & lockdown) under the epidemic of COVID-19, but the loss can be controlled by taking alternating strategies based on time-based and result-based switching scheme – a phenomenon attributed to Parrondo's paradox.
- Reviewed the development and progress of fractal characteristics in the network systematically, including network covering, various concepts of dimensions, and their real-world applications.

Finding Important Properties of Nodes and Communities in Complex Networks June 2019–Sep. 2019
University of Electronic Science and Technology of China, Advisor: Professor Yong Deng

- Combined the internal factors and external factors of community to measure the vulnerability of each community and improved the recognition accuracy in real-world network applications via the entropy-based method.
- Collected the personal opinion of each expert by 2-order additive fuzzy measure and identified the most important expert after constructing the expert network based on their relationships, thereby solving group decision-making problems without sufficient information.

Exploring the Properties of Networks by Different Dimension May 2016–June 2019
Northwestern Polytechnical University, Advisor: Professor Wen Jiang & Yong Deng

- Promoted information dimension and Rényi dimension into weighted complex networks and explored the fractal and self-similarity properties of complex networks.
- Developed several recognition models based on the dimension of the network to solve practical problems, such as measuring the complexity and vulnerability of airline networks, as well as identifying the similarity and importance of individuals.

Evaluating Vulnerability of Networks Based on Fuzzy Fractal Dimension May 2017–May 2018
Team Leader, National Undergraduate Training Programs for Innovation and Entrepreneurship (￥20, 000)

An Autonomous Landing Scheme for Cargo Drone Based on Computer Vision May 2017–May 2018
Researcher, National Undergraduate Training Programs for Innovation and Entrepreneurship (￥20, 000)

Exploring the Composition and Future Scalability of Smart House Jan. 2019–Feb. 2019
Australian National University, Summer Session Course Project

AWARDS & HONORS

- Decision and Cognitive Sciences Research Centre Support Fund 2022
- ESRC funded integrated PhD and MSc in Data Analytics and Society Scholarship 2021
  Only one international PhD scholarship in 2021.

The **first** scholarship for Chinese PhD student in this CDT Program.

- Singapore University of Technology and Design MEng (Research) Fellowship 2020
- The "Challenge Cup" National Undergraduate Extracurricular Academic Science and Technology Contest: **Special First Prize** (**Top 4%**) 2019

  Highest award in the field of information science in natural science papers.

  The "Challenge Cup" is the **highest academic science and technology competition** in China.
- The "Challenge Cup" National Undergraduate Extracurricular Academic Science and Technology Contest in Shaanxi Area: **Outstanding Winner** (**TOP 5%**) 2019

  Best grade in the field of natural science papers in Northwestern Polytechnical University **so far**.
- **Outstanding Student Pacemaker in Northwestern Polytechnical University** (**TOP 0.1%**) 2018

  Only **10 undergraduates** are awarded among more than 12,000 undergraduates in NWPU.
- **China National Scholarship** (**Top 1%**) 2018

  Highest honor for students' year achievement in the School of Electronics and Information.
- Global Mathematical Contest in Modeling (MCM/ICM): **Meritorious Winner** 2018
- Chinese Undergraduate Mathematical Contest in Modeling in Shaanxi Area: **First Prize** 2017
- Second Prize scholarship of AVIC Optronics Institute (**TOP 5%**) 2017
- First Prize scholarship of Northwestern Polytechnical University (**TOP 10%**) 2017
- Global Mathematical Contest in Modeling (MCM/ICM): **Honorable Mention** 2017

ACADEMIC SERVICE

- Conference talk at 26th International Conference on Multiple Criteria Decision Making, Portsmouth, UK.
- Talk at the Oberseminar Dynamics Seminar in Technical University of Munich (Online), Munich, Germany.
- Conference talks at 9th China Information Fusion Conference, Taiyuan, China.
- Talk at Ao Xiang Students' Lecture in NWPU, Xi'an, China.
- Teaching assistant of 10.018 (Modelling Space and Systems), SUTD.
- Independent reviewer for:
  - *Communications Physics*
  - *Chaos: An Interdisciplinary Journal of Nonlinear Science*
  - *IEEE Transactions on Signal and Information Processing over Networks*
  - *International Journal of Intelligent Systems*
  - *Journal of Computational Science*
  - *Mathematics*
  - *Physica A: Statistical Mechanics and Its Applications*

SOCIAL ACTIVITIES

- **Volunteer** | International Conference on Intelligent Unmanned System 2018
- **Volunteer** | Northwestern Polytechnical University 80th Anniversary 2018

SKILLS

Technical: MATLAB, Python, C language, LaTeX, Gephi, Origin.

Languages: Mandarin, English.

REFERENCES

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Last updated: July 5, 2022